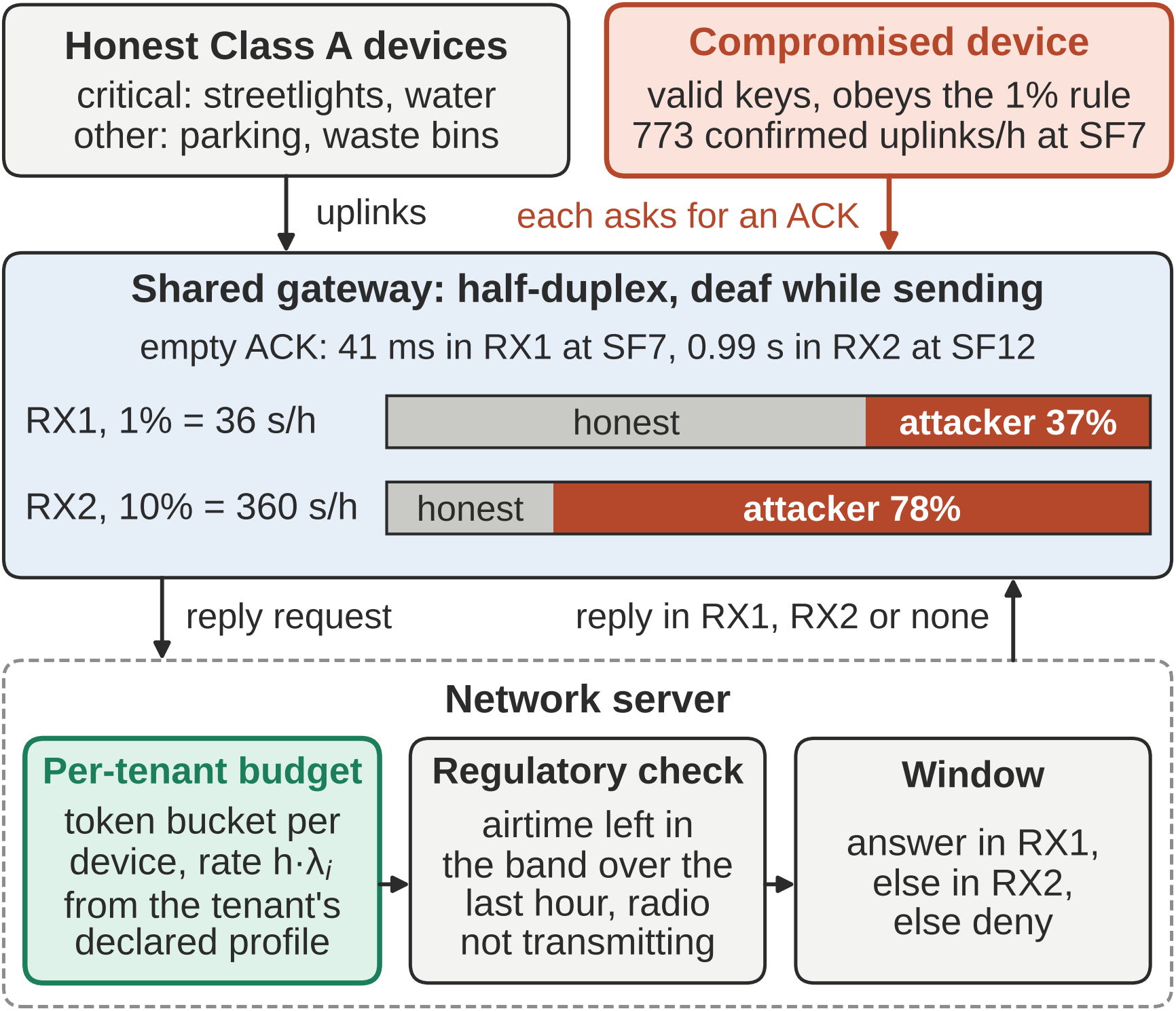
Honest Class A devices
critical: streetlights, water
other: parking, waste bins
Compromised device
valid keys, obeys the 1% rule
773 confirmed uplinks/h at SF7
uplinks
each asks for an ACK
Shared gateway: half-duplex, deaf while sending
empty ACK: 41 ms in RX1 at SF7, 0.99 s in RX2 at SF12
RX1, 1% = 36 s/h
honest
attacker 37%
RX2, 10% = 360 s/h
honest
attacker 78%
reply request
reply in RX1, RX2 or none
Network server
Per-tenant budget
token bucket per device, rate h·λi from the tenant's declared profile
Regulatory check
airtime left in the band over the last hour, radio not transmitting
Window
answer in RX1, else in RX2, else deny